ManyDaughters Pre-analysis Plan

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1 Hypotheses

1. Having daughters leads to stronger support for gender equality (as measured by attitudes).
2. Having daughters leads to more left-leaning political preferences.
3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

2 Analysis

We interpret all hypotheses as comparing respondents with daughters to comparing respondents with sons. For each hypothesis, we look at the outcomes in the year after the first child was born. Since the sex of a child is random, the cleanest comparison is to compare respondents whose first child is a girl to those whose first child is a boy, shortly after their child was born. For this, we create a variable called `first_daughter` taking on the value 1 if the first child is a girl, and 0 if it is a boy. We also create a variable denoting the birth year of the first child (`first_byear`). We include all respondents with children, and we do not exlude any observations from the main dataset. However, each hypothesis has a slightly different sample defined by the survey years in which the outcome variables are available, and they only include respondents whose first child was born in the year before one of the survey years.

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2.1 Hypothesis 1

For Hypothesis 1, we create an index of three variables using principal component analysis. The index is created using the first component weights of the three variables. The three variables are

- `pl0298_v1` in 2012 and `pl0298_v2` in 2018: "Children under the age of 6 suffer if their mother is working."
- `pl0302_v1` in 2012 and `pl0302_v2` in 2018: "Children under the age of 3 suffer if their mother is working."
- `pl0308_v1` in 2012 and `pl0308_v2` in 2018: "Best if man and woman work the same amount so they can share the responsibility."

We use these variables because they have the most nonmissing values and the greatest variation in the answers in the years when they are available. Since the answers in 2012 are coded on a 1 to 4 scale, while in 2018 on a 1 to 7 scale, we first standardize all variables and then create the gender attitude index using the data from both years.

The regression we run is the following:

$$\text{gender_attitudes}_i = \alpha_i + \beta_1 \text{first_daughter}_i + \beta_2 \text{age}_i + \beta_3 \text{survey_year}_i + \epsilon_i \tag{1}$$

The outcome variable is the gender attitude index measured the year after the first child was born. It is greater the more equal the respondent's attitudes are. The treatment variable is whether the first child is a daughter (vs. a son). We include the age of the respondent in all of the regressions, because there is some evidence that older fathers have a higher chance to have daughters (e.g., Jacobsen et al., 1999; Sánchez-Barricarte, 2023; Ha et al., 2024). We also include survey year fixed effects, to account for potential differences between the years when the questions were asked. We cluster the standard errors on the household level, because the respondents were sampled on household level, and answers of respondents from the same household might be correlated.

The coefficient of interest is β_1, the effect of the first child being a girl. Since the outcome variable takes on a higher value if the respondent's attitudes are more equal, a significantly positive β_1 would mean the result is in line with the hypothesis. We use a t-test and a 5 percent significance level.

2.2 Hypothesis 2

For hypothesis 2, we use one variable as the outcome, `plh0004`, which is the political orientation of the respondent on a 0 to 10 scale. 0 means completely left and 10 means completely right. This

question was asked in 2005, 2009, 2014 and 2019, so we use data from respondents whose first child was born in the year before one of these survey years.

The regression we run is the following:

$$\text{political_attitude}_i = \alpha_i + \beta_1 \text{first_daughter}_i + \beta_2 \text{age}_i + \beta_3 \text{survey_year}_i + \epsilon_i \tag{2}$$

The control variables are the same as in Hypothesis 1, and the outcome variable is the political attitude scale measured in the year after the first child was born. The standard errors are clustered on the household level. If β_1 is significantly negative, then the result is in line with the hypothesis. We use a t-test and a 5 percent significance level.

2.3 Hypothesis 3

For Hypothesis 3, we use the time use questions that are available in every odd year or every year. We define gender equality in the household by the equality of the hours spent on nonpaid work by the mother and the father. The variables we use for nonpaid work are the following:

- `pli0011`: Hours on running errands on Sunday
- `pli0012_h`: Hours on housework on Saturday
- `pli0016_h`: Hours on housework on Sunday
- `pli0019_h`: Hours on childcare on Saturday
- `pli0022_h`: Hours on childcare on Sunday
- `pli0031_h`: Hours on reparation on Saturday
- `pli0040`: Hours running errands on workdays
- `pli0043_h`: Hours on housework on workdays
- `pli0044_h`: Hours on childcare on workdays
- `pli0046`: Hours on caring for persons on workdays
- `pli0049_h`: Hours on reparations on workdays
- `pli0055`: Hours supporting persons in care on Saturday
- `pli0057`: Hours supporting persons in care on Sunday

We add up the hours to get weekly hours (for weekday variables we use five times the hours reported), cap the variable at 168 hours, and calculate the Gini coefficient between partners i and j in the same household the following way:

$$\text{nonpaid_gini} = \frac{|\text{total_nonpaid_hours}_i - \text{total_nonpaid_hours}_j|}{\text{total_nonpaid_hours}_i + \text{total_nonpaid_hours}_j} \tag{3}$$

The Gini coefficient is 0 for full equality and 1 for full inequality. In the end, we have data from every odd year between 2001 and 2019 and from 2012. Again, we use the responses from the year after the first child was born.

The regression we run is the following:

$$\text{nonpaid_gini}_h = \alpha_h + \beta_1 \text{first_daughter}_h + \beta_2 \text{age_father}_h + \beta_3 \text{survey_year}_h + \epsilon_h \tag{4}$$

One observation is now one household. We control for the survey year and the age of the father. If β_1 is significantly negative, the result is in line with the hypothesis. We use a t-test and a 5 percent significance level.

2.4 Hypothesis 4

For Hypothesis 4, we use the same dataset as for Hypothesis 1, but the regression we run is different:

$$\text{gender_attitudes}_i = \alpha_i + \beta_1 \text{first_daughter}_i + \beta_2 \text{female}_i + \beta_3 \text{first_daughter}_i \times \text{female}_i + \beta_4 \text{age}_i \tag{5}$$

$$+ \beta_5 \text{survey_year}_i + \epsilon_i \tag{6}$$

Here the standard errors are clustered on the household level. female_i shows whether the respondent is female, and the coefficient of interest is β_3. A significantly negative β_3 is in line with the hypothesis. We use a t-test and a 5 percent significance level.

References

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